Nama : Anjaanie Mafaaza Sahilah

NIM = 2510514023

Parameter Personalisasi :

$N = 23,\ a = 2,\ b = 3,\ K = 6,\ \theta_0 = 75^\circ,\ \alpha = 0{,}004$

Soal 2

dik: matriks $A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix}$ $\quad a = 2,\ b = 3,\ N = 23$

Jawab : matriks $A = \begin{bmatrix} 2+1 & 3 & 2 \\ 1 & 2+2 & 3 \\ 3 & 1 & 2+3 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 2 \\ 1 & 4 & 3 \\ 3 & 1 & 5 \end{bmatrix}$

a). determinan A (ekspansi baris pertama)

$\det(A) = 3M_{11} - 3M_{12} + 2M_{13}$

$M_{11} = \begin{vmatrix} 4 & 3 \\ 1 & 5 \end{vmatrix} = 20 - 3 = 17$

$M_{12} = \begin{vmatrix} 1 & 3 \\ 3 & 5 \end{vmatrix} = 5 - 9 = -4$

$M_{13} = \begin{vmatrix} 1 & 4 \\ 3 & 1 \end{vmatrix} = 1 - 12 = -11$

$\det(A) = 3(17) - 3(-4) + 2(-11) = 51 + 12 - 22 = 41$ //

b). Matriks minor (9 elemen)

Baris 1

$M_{11} = 17,\ M_{12} = -4,\ M_{13} = -11$

Baris 2

$M_{21} = \begin{vmatrix} 3 & 2 \\ 1 & 5 \end{vmatrix} = 15 - 2 = 13$

$M_{22} = \begin{vmatrix} 3 & 2 \\ 3 & 5 \end{vmatrix} = 15 - 6 = 9$

$M_{23} = \begin{vmatrix} 3 & 3 \\ 3 & 1 \end{vmatrix} = 3 - 9 = -6$

Baris 3

$M_{31} = \begin{vmatrix} 3 & 2 \\ 4 & 3 \end{vmatrix} = 9 - 8 = 1$

$M_{32} = \begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix} = 9 - 2 = 7$

$M_{33} = \begin{vmatrix} 3 & 3 \\ 1 & 4 \end{vmatrix} = 12 - 3 = 9$

Matriks Minor

$M = \begin{bmatrix} 17 & -4 & -11 \\ 13 & 9 & -6 \\ 1 & 7 & 9 \end{bmatrix}$

c). Kofaktor, Adjoin, Invers

* Matriks kofaktor (+ - +)

$C = \begin{bmatrix} 17 & 4 & -11 \\ -13 & 9 & 6 \\ 1 & -7 & 9 \end{bmatrix}$

* Adjoin (transpose)

$Adj(A) = \begin{bmatrix} 17 & -13 & 1 \\ 4 & 9 & -7 \\ -11 & 6 & 9 \end{bmatrix}$

* Invers

$A^{-1} = \frac{1}{41} \begin{bmatrix} 17 & -13 & 1 \\ 4 & 9 & -7 \\ -11 & 6 & 9 \end{bmatrix}$

PAPERLINE

Dipindai dengan CamScanner

No.

Date :

e). Enkripsi dan Dekripsi

Matriks $P = \begin{bmatrix} 120 & (50+N) & 200 \\ 80 & a+2 & (10+N) \\ (30+N) & 90 & 170 \end{bmatrix}$

$$P = \begin{bmatrix} 120 & 50+23 & 200 \\ 80 & 2+2 & 10+23 \\ 30+23 & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 73 & 200 \\ 80 & 4 & 33 \\ 53 & 90 & 170 \end{bmatrix}$$

Enkripsi : $E = A \cdot P = \begin{bmatrix} 3 & 3 & 2 \\ 1 & 4 & 3 \\ 3 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 120 & 73 & 200 \\ 80 & 4 & 33 \\ 53 & 90 & 170 \end{bmatrix}$

Baris 1

$E_{11} = 3(120) + 3(80) + 2(53) = 360 + 240 + 106 = 706$

$E_{12} = 3(73) + 3(4) + 2(90) = 219 + 12 + 180 = 411$

$E_{13} = 3(200) + 3(33) + 2(170) = 600 + 99 + 340 = 1039$

Baris 2

$E_{21} = 1(120) + 4(80) + 3(53) = 120 + 320 + 159 = 599$

$E_{22} = 1(73) + 4(4) + 3(90) = 73 + 16 + 270 = 359$

$E_{23} = 1(200) + 4(33) + 3(170) = 200 + 132 + 510 = 842$

Baris 3

$E_{31} = 3(120) + 1(80) + 5(53) = 360 + 80 + 265 = 705$

$E_{32} = 3(73) + 1(4) + 5(90) = 219 + 4 + 450 = 673$

$E_{33} = 3(200) + 1(33) + 5(170) = 600 + 33 + 850 = 1483$

matriks $E = \begin{bmatrix} 706 & 411 & 1039 \\ 599 & 359 & 842 \\ 705 & 673 & 1483 \end{bmatrix}$

$P' = A^{-1} \cdot E = \frac{1}{41}(Adj(A) \cdot E) = \begin{bmatrix} 17 & -13 & 1 \\ 4 & 9 & -7 \\ -11 & 6 & 9 \end{bmatrix} \cdot \begin{bmatrix} 706 & 411 & 1039 \\ 599 & 359 & 842 \\ 705 & 673 & 1483 \end{bmatrix}$

Baris 1

$(1,1) = 17(706) + (-13)(599) + 1(705) = 12002 - 7787 + 705 = 4920,\ P'_{11} = \frac{4920}{41} = 120$

$(1,2) = 17(411) + (-13)(459) + 1(673) = 6987 - 4667 + 673 = 2993,\ P'_{12} = \frac{2993}{41} = 73$

$(1,3) = 17(1039) + (-13)(842) + 1(1483) = 17663 - 10946 + 1483 = 8200,\ P'_{13} = \frac{8200}{41} = 200$

Baris 2

$(2,1) = 4(706) + 9(599) + (-7)(705) = 2824 + 5391 - 4935 = 3280,\ P'_{21} = \frac{3280}{41} = 80$

$(2,2) = 4(411) + 9(459) + (-7)(673) = 1644 + 3231 - 4711 = 164,\ P'_{22} = \frac{164}{41} = 4$

$(2,3) = 4(1039) + 9(842) + (-7)(1483) = 4156 + 7578 - 10381 = 1353,\ P'_{23} = \frac{1353}{41} = 33$

Baris 3

$(3,1) = (-11)(706) + 6(599) + 9(705) = -7766 + 3594 + 6345 = 2173,\ P'_{31} = \frac{2173}{41} = 53$

$(3,2) = (-11)(411) + 6(459) + 9(673) = -4521 + 2754 + 6057 = 3690,\ P'_{32} = \frac{3690}{41} = 90$

$(3,3) = (-11)(1039) + 6(842) + 9(1483) = -11429 + 5052 + 13347 = 6970,\ P'_{33} = \frac{6970}{41} = 170$

$P' = \begin{bmatrix} 120 & 73 & 200 \\ 80 & 4 & 33 \\ 53 & 90 & 170 \end{bmatrix}$, $P = P' \rightarrow$ Terbukti benar.

PAPERLINE

Dipindai dengan CamScanner

No.

Date :

Jika det(A) = 0, maka :

1. Matriks tidak memiliki invers
2. Proses deskripsi $P = A^{-1} E$ tidak bisa dilakukan
3. Informasi dalam matriks terenkripsi tidak dapat dikembalikan ke bentuk asli
4. Dalam konteks enkripsi, sistem menjadi tidak valid karena kehilangan kemampuan utk membalikkan proses. Artinya, data bisa dienkripsi tapi tidak bisa dideskripsi/hilang.

Dipindai dengan CamScanner